Multiplying Fractions and Mixed Numbers

Multiplying a fraction by a fraction, a fraction by a whole number, mixed numbers by rewriting them as improper fractions, and finding a missing factor.

- Show the multiplication you set up, not just the final number.
- Give every answer in simplest form. Improper fractions are fine.
- Rewrite a mixed number as an improper fraction *before* you multiply.

1. Multiply. Simplify your answer.

$$\frac{2}{3} \times \frac{3}{5}$$

Answer: ________________

2. Multiply. Simplify your answer.

$$\frac{3}{4} \times \frac{1}{2}$$

Answer: ________________

3. Multiply a fraction by a whole number. Simplify your answer.

$$\frac{5}{6} \times 4$$

Answer: ________________

4. Multiply. Look for a common factor you can cancel before you multiply.

$$\frac{3}{8} \times \frac{2}{9}$$

Answer: ________________

5. Rewrite the mixed number as an improper fraction first, then multiply.

$$2\frac{1}{2} \times \frac{3}{5}$$

Answer: ____________________

6. Find the missing factor. What number times $\frac{3}{4}$ gives $\frac{9}{20}$?

$$\frac{3}{4} \times x = \frac{9}{20}$$

Answer: ____________________

7. Multiply two mixed numbers. Simplify your answer.

$$1\frac{1}{3} \times 2\frac{1}{4}$$

Answer: __________________

8. A batch of trail mix uses $\frac{3}{8}$ of a cup of raisins. Ravi makes 12 batches. How many cups of raisins does he use in all?

Answer: __________________

9. Multiply. Cancel common factors before multiplying — the numbers stay small.

$$\frac{7}{10} \times \frac{5}{14}$$

Answer: ________________

10. A rectangular doormat measures $2\frac{3}{4}$ feet by $1\frac{1}{2}$ feet. Find its area. Give the answer as a decimal number of square feet.

Answer: ________________ ft^2

11. Find the missing factor.

$$x \times 2\frac{1}{2} = 4$$

Answer: ________________

12. A trail is $4\frac{1}{2}$ kilometres long. On Saturday Maya walks $\dfrac{2}{3}$ of the whole trail. On Sunday she walks $\dfrac{1}{2}$ of the part that was left. How far does she walk on Sunday?

Answer: ________________ km